

1.2 Defined and Undefined Terms – Part 2

Lesson Introduction

 Benchmark	MA.912.GR.2.2 Identify transformations that do or do not preserve distance.		 Learning Targets	- I can understand the three undefined terms of geometry. - I can use rigid motions to define a straight angle. - I can experiment with rigid motions on a plane.	
 Benchmark Coherence	Building On			Working Toward	
	MA.8.GR.2.1 Given a preimage and image generated by a single transformation, identify the transformation that describes the relationship.			N/A	
 Essential Questions	- What is a definition? - What is meant by the phrase “undefined term”? - What are the properties of the three undefined terms of Geometry: point, line, and plane?				
 Lesson Narrative	This is the second lesson of a two-part lesson. Students will use patty paper to explore the rotation of a line segment and determine that a straight line has a degree measure of 180°. The undefined terms of Geometry are introduced along with their properties.				
 Component Summary	1.2.1 Warm-Up (~5 min.)	Justify the choice of two amounts (<i>SE p. 10</i>)	1.2.3 Bring It Together (10 min.)	Identify the three undefined terms of Geometry (<i>SE p. 11</i>)	
	1.2.2 Exploration (25 min.)	Explore the rotation of a line segment (<i>SE p. 10</i>)	1.2.4 Wrap-Up (~5 min.)	Students rate their level of confidence (<i>SE p. 12</i>)	
 Resources	Glossary		Materials		Additional Preparation
	<u>Key Terms:</u> - Straight Angle - Definition <u>Additional Terms:</u> - Undefined Term - Point - Line - Plane		- Patty Paper (Optional) Sticky Notes or Notecards		Consider using point, line, and plane cards or white boards as response tools for debriefing 1.2.3 Bring It Together question 2.

 Teacher Tips	Common Misconceptions • Students may not understand why a term is undefined. • Students may confuse the properties of a point, line, and plane.	Things to Consider • Students may struggle with the directions in 1.2.2 Exploration. Consider having samples that can be used for demonstration if needed. • Consider having students tape or paste their patty paper into their workbook.
	Literacy and Language Routines Poll the Class Consider displaying the prompts found in the table from 1.2.3 Bring It Together one at a time. Have students indicate their choice of answer either by holding up their answer on paper, a whiteboard, or using technology devices. Display the results of initial reports before students are asked to defend their choice.	Mathematical Thinking and Reasoning (MTR) Standard MA.K12.MTR.1.1: Participate in Effortful Learning In 1.2.2 Exploration, teachers should encourage students to participate actively in effortful learning to persevere when they are unsure of steps when using patty paper to construct different angles. Students should stay engaged and maintain a positive mindset while working on 1.2.2 Exploration. Teachers can help students support each other by providing sentence stems.
	 Differentiate Instruction	
Lesson Notes:		

1.2.1 Warm-Up: Would You Rather?

Component Overview: Students choose and justify their reasoning on whether they would rather have a stack of pennies or a stack of quarters to match their weight or height.



1.2.1 Warm-Up: Would You Rather?

1. Would you rather have pennies to match your weight or quarters to match your height?
Answers will vary.
2. Explain your reasoning.
Answers will vary.



Warm-Ups often provide all students with an entry-point and a safe place to be successful. Allow students to answer without the fear of being wrong. Consider providing the US Mint's coin specifications for students who want more details. Consider using 1.2.1 Warm-Up to further establish class norms and routines.

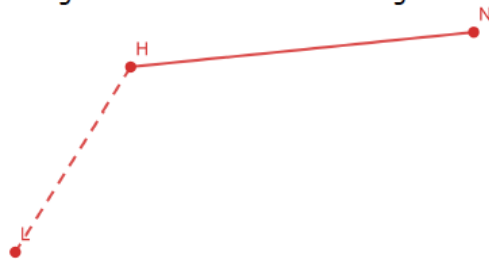
1.2.2 Exploration: Definitions

Component Overview: Students will explore rotating a line segment using patty paper.

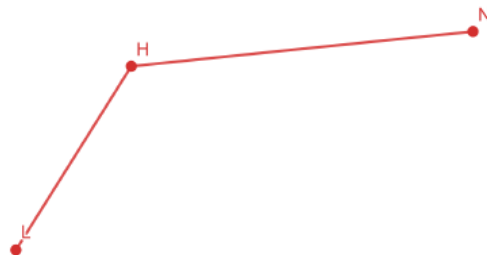


1.2.2 Exploration: Definitions

- Use two sheets of patty paper to complete the following.
 - Draw $\overline{HN}$ on one piece of patty paper.
 - Draw $\overline{KL}$ on the other piece of patty paper.
 - Lay the piece of patty paper with $\overline{HN}$ drawn on it on top of the piece with $\overline{KL}$ drawn on it. Place the papers so that the points H and B are on top of each other and the line segments coincide.
 - Place your writing utensil on endpoint H .
 - Rotate the top sheet of patty paper until the two line segments make an obtuse angle.



- Draw the obtuse angle you created below.



- When you rotated the top sheet, did the line segment change in size?

No



Consider beginning the lesson with a short overview of things that students remember from Lesson 1. Allow students to use their workbook to share ideas. Consider using a Think-Pair-Share routine by prompting students to write down 3 things they learned from Lesson 1. Have students share their thinking in a low-stakes way with one partner. Then each pair can share one thing with the class. As you circulate eavesdrop on the partner conversations so that you can purposefully select students to share with the class. This review will be helpful for students during question 2 of 1.2.3 Bring It Together.



To support the opportunity for students to expand their vocabulary, ask the following questions:

- When is the angle acute?
- At what position is a right angle formed?
- Where is the angle obtuse?

1.2.2 Exploration: Definitions (continued)

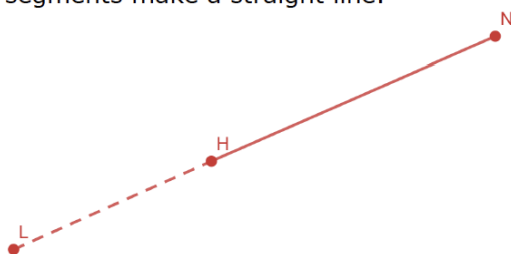
4. When you rotated the top sheet, did the line segment change in position?

Yes

5. Which part(s) of the obtuse angle has two points?

Each side of the obtuse angle, segment HN and segment HL , have two points.

6. Rotate the top sheet of patty paper until the two line segments make a straight line.



7. Name the resulting line segment using the endpoints.

The resulting line segment is $\overline{NL}$.

8. Work with your partner to estimate the number of degrees you rotated the top sheet of patty paper from its original position on top of line segment BL .

Answers will vary. I estimate that I rotated the shape about 180° .



A **straight angle** is an angle measuring exactly 180° .

9. Based on the rotation and the definition of a straight angle, what "fact" about a line can be added to the "Facts about Lines" in the previous lesson?

A line measures 180° .



Students are experiencing transformations that preserve distance. This concept of preserving distance is not formally established but is used to build student understanding of definitions.

1.2.3 Bring It Together: Undefined Terms

Component Overview: Students are introduced to what a definition is and why there are undefined terms in Geometry. Then, the three undefined terms are matched to their properties.



1.2.3 Bring It Together: Undefined Terms



A **definition** is an exact statement, written formally, of the meaning or description of a word.

There are three terms in geometry that are considered “undefined terms.”

A term is considered undefined if, when explaining the term, the explanation relies on the term itself or a close synonym. This is sometimes called a circular definition.

1. In the previous lesson you wrote a definition for the term “line.” Review your definition and determine if your definition is considered circular.

Answers will vary. I would consider my definition circular.

Although we have “facts about lines,” “line” is an undefined term in geometry. The other two undefined terms in geometry are “point” and “plane.” Look back through the previous lesson to notice how often these terms were used without providing a definition.



In Geometry, line is considered an undefined term. Other courses may have provided a definition based on the properties of a line but those are not considered to be accepted definitions as part of this course.

Consider having students write their definitions on sticky notes or notecards for you to use as examples to show why the definitions are circular. Question 1 is a discussion opportunity. Students do not need to write an explanation here, but a sample response is provided for reference.

1.2.3 Bring It Together: Undefined Terms (continued)

2. Work with your partner to label each description as a line, plane, or point.

Description	Undefined Term
has two dimensions	☐ Line ☐ Point ☒ Plane
has no length, no width, and no height	☐ Line ☒ Point ☐ Plane
is often drawn as a four-sided figure	☐ Line ☐ Point ☒ Plane
has infinite length, no width, no height	☒ Line ☐ Point ☐ Plane
named by an ordered pair in the coordinate plane	☐ Line ☒ Point ☐ Plane
indicates a location or position	☐ Line ☒ Point ☐ Plane
has one dimension	☒ Line ☐ Point ☐ Plane
has infinite length, infinite width, and zero height	☐ Line ☐ Point ☒ Plane
has no dimension or actual size	☐ Line ☒ Point ☐ Plane



Use the literacy routine “Poll the Class” to gauge student understanding of the three undefined terms of Geometry. Display the prompts found in the table one at a time and have students indicate their choice of answer either by holding up their answer on paper or a whiteboard or using technology devices. Display the results of initial reports before students are asked to defend their choice.

1.2.4 Wrap-Up: Reflection

Component Overview: Students reflect on their confidence with the new material by providing a rating of 1 – 4.



1.2.4 Wrap-Up: Reflection

1. Rate yourself based on what you understood in today's lesson.

Answers will vary.

Skill	1 – I am just starting. 2 – I have some skills. 3 – I am almost there. 4 – I have it!
I can understand the three undefined terms of geometry.	
I can define a straight angle.	

2. Rate yourself based on how you feel about the following statements.

Answers will vary.

Statement	1 – I struggle with this. 2 – I am ready to grow in this area. 3 – I am almost there. 4 – That's me!
I am a student learner.	
I am a mathematician.	



Cruising Clipboard

Walk around the room and quickly make note of the sum of the values for the first box.

- When most students are having sums of ten or higher, students are confident.
- If students are closer to a six to nine range, they may benefit from another opportunity to solidify these definitions.
- Any time scores are five or under, consider reteaching this lesson, or intentionally integrating these definitions early in the next few lessons.